



Implantable drug delivery device

Researchers have developed a refillable, implantable device that can deliver drugs to your heart <https://digit.in/jul18-78-1>



Planet where a year is 19.5 days

Indian scientists have found an exoplanet, 600 light years away from Earth, where a year is just 19.5 days <https://digit.in/jul18-78-2>

As good as new!

Can we heal like Wolverine or Deadpool in real life?

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It happens every time. When we see Wolverine, Deadpool, Dorian Grey or Nagraj heal them-

selves in no time, we cannot help but wish us regular humans had the same powers too. Imagine, any cut, any bruise healed in seconds - even entire limbs growing back from stumps! Sadly, much like most fictional powers we've desired for ages, accelerated healing has mostly been the stuff of fiction. Or so we would believe. Turns out, theoretically, accelerated healing is not far fetched at all. In fact, it exists out there in nature! And if recent research is to be believed, we could be on the verge of unlocking it for ourselves. But first, a little background - about salamanders.

SUPER-SALAMANDER

It isn't hard to observe that we already have differential rates of healing, even within the human species. People suffering from immunodeficiencies take much longer to heal even from small cuts and wounds than a regular person. On the other hand, people who work out regularly, or even have higher levels of certain hormones, heal faster from bruises, impacts and abrasions. Looking beyond humans, though, brings up a whole different scene - one that is way more accelerated and capable.

Some species in nature have unlimited regenerative capabilities.



Axolotls - these cute salamanders are nature's super healers

The group of amphibians known as salamanders, that includes newts and axolotls can regrow entire organs after an amputation and even replace crucial organs such as the brain or the heart! Makes you think, if some animals can do this, why can't all? That's because regeneration isn't a good survival strategy. It takes longer than, say, the immediate closure of an open wound through scarring - which is essential to avoid infections or bleeding out. Overall, scarring - the quick, rough way of dealing with wounds - wins most of the time.

SCARS? WHAT SCARS?

From where we stand today, the first goal in the direction of accelerated healing is less scar formation. Scar tissues are different from regular tissues in the sense that they are formed by side-by-side arrangements of collagen fibres as compared to the weaves in normal tissues. Think of it like stitches holding a fabric together. This is easy to do but leaves a mark, and is not as

strong as regular tissue.

To replace scar formation, scientists have come up with several approaches to speed up the regenerative process. We must understand that this happens anyways for most scars, except ones that result in a significant loss of tissues. The process involves initial healing through clotting, then further increase in something known as growth factors (GFs). These growth factors, if increased in quantity and deployed at the correct places, can actually make wounds heal and close faster without leaving a scar.

THE METHODS

The use of platelet rich plasma and platelet rich fibrin, both derived from whole blood, have proven to be beneficial in accelerating healing. In fact, they are used in some branches of medicine, including dentistry. Recently, in a few studies, they've even proven their efficacy towards treating ulcers. Atrophic scars like the ones left behind by acne and measles can be